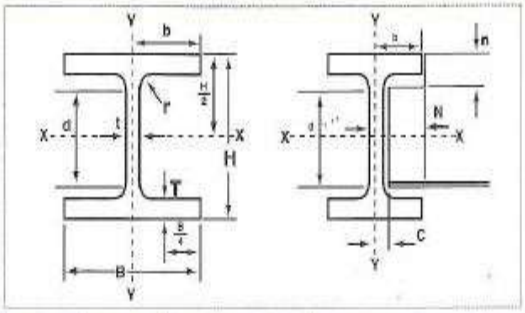
Dimensions and Properties

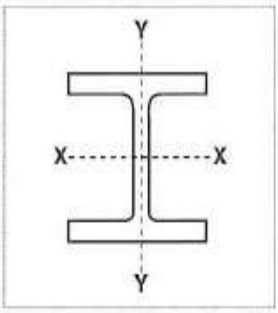


4a (i) Beams (Imperial Units) - Part I

	Designation Size		Unit Weight		Height Of Beam	Width Of Beam	Thickness		Root Radius	Depth Between Fillets	Area Of Section	Ratios For Local Buckling		Dimensions For Detailing		
							Flange	Web				Flange	Web	End Clearance	Notch	
					H	B	T	t	r	d	A	b/T	d/t	C	N	n
	in	(mm)	lb/ft	kg/m	mm	mm	mm	mm	mm	mm	cm²			mm	mm	mm
W40	40 x 12	(1016 x 305)	149	221.7	970.3	300.0	21.1	16.0	30.0	868.1	283	7.11	54.3	10	152	51
			167	248.5	980.2	300.0	26.0	16.5	30.0	868.2	317	5.77	52.6	10	152	56
			183	272.3	990.1	300.0	31.0	16.5	30.0	868.1	347	4.84	52.6	10	152	61
			211	314.0	1000.0	300.0	35.9	19.1	30.0	868.2	400	4.18	45.5	12	150	66
			235	349.7	1008.1	302.0	40.0	21.1	30.0	868.1	445	3.78	41.1	13	150	70
			264	392.9	1016.0	303.0	43.9	24.4	30.0	868.2	500	3.45	35.6	14	149	74
			278	413.7	1020.0	304.0	46.0	26.0	30.0	868.0	529	3.30	33.4	15	149	76
			294	437.5	1025.9	305.4	49.0	26.9	30.0	867.9	557	3.12	32.3	15	149	79
			327	486.6	1036.1	308.5	54.1	30.0	30.0	867.9	620	2.85	28.9	17	149	84
			331	492.6	1036.0	309.0	54.0	31.0	30.0	868.0	629	2.86	28.0	18	149	84
			359	534.2	1046.0	311.4	58.9	33.0	30.0	868.2	681	2.64	26.3	19	149	89
			392	583.4	1056.0	314.0	64.0	36.0	30.0	868.0	744	2.45	24.1	20	149	94
	40 x 16	(1016 x 406)	199	296.1	982.0	400.0	27.1	16.5	30.0	868.0	377	12.12	32.0	10	202	57
			215	320.0	990.0	400.0	31.0	16.5	30.0	868.0	409	12.12	28.0	10	202	61
			249	370.6	1000.0	400.0	36.1	19.0	30.0	868.0	472	10.53	24.0	12	201	66
			277	412.2	1008.0	402.0	40.0	21.1	30.0	868.0	524	9.53	21.7	13	200	70
			297	442.0	1012.0	402.0	41.9	23.6	30.0	868.2	564	8.52	20.7	14	199	72
			324	482.2	1020.0	404.0	46.0	25.4	30.0	868.0	615	7.95	18.9	15	199	76
			362	538.7	1030.0	407.0	51.1	28.4	30.0	867.8	687	7.17	17.0	16	199	81
			397	590.8	1040.0	409.0	55.9	31.0	30.0	868.2	753	6.60	15.5	18	199	86
			431	641.4	1048.0	412.0	60.0	34.0	30.0	868.0	818	6.06	14.5	19	199	90
			503	748.5	1068.0	417.0	70.0	39.0	30.0	868.0	953	5.35	12.4	22	199	100
			593	882.5	1092.0	424.0	82.0	45.5	30.0	868.0	1125	4.66	10.6	25	199	112
W44	44 x 16	(1118 x 406)	230	342.3	1090.0	400.0	31.0	18.0	20.0	988.0	437	11.11	31.9	11	201	51
			262	389.9	1100.0	400.0	36.0	20.0	20.0	988.0	497	10.00	27.4	12	200	56
			290	431.6	1108.0	402.0	40.0	22.0	20.0	988.0	551	9.14	24.7	13	200	60
			335	498.5	1118.0	405.0	45.0	26.0	20.0	988.0	635	7.79	22.0	15	200	65

Note: Calculated based on 1 kg = 2.2046 lb and 1 m = 3.2808 feet.
*Please contact us for further clarification.

Dimensions and Properties



4a (i) Beams (Imperial Units) - Part II

Unit Weight		Surface Area per Metre	Second Moment Of Area		Radius Of Gyration		Elastic Modulus		Plastic Modulus		Buckling Parameter	Torsional Index	Warping Constant	Torsional Constant	Designation Size	
			Axis x-x	Axis y-y	Axis x-x	Axis y-y	Axis x-x	Axis y-y	Axis x-x	Axis y-y						
lb/ft	kg/m	m^2	cm^4	cm^4	cm	cm	cm^3	cm^3	cm^3	cm^3	u	x	H	J	in	
149	221.74	3.06	408000	9553	38.0	5.81	8409	637	9807	1020	0.850	45.7	21.5	390	**40 x 12**	**W40**
167	248.52	3.08	481300	11760	39.0	6.09	9821	784	11350	1245	0.861	39.9	26.8	582		
183	272.33	3.10	554000	14010	40.0	6.36	11190	934	12830	1470	0.873	35.0	32.2	835		
211	314.00	3.11	644200	16240	40.1	6.37	12880	1083	14850	1713	0.872	30.7	37.7	1264		
235	349.72	3.13	723100	18470	40.3	6.44	14350	1223	16590	1941	0.872	27.9	43.3	1718		
264	392.87	3.14	807700	20500	40.2	6.40	15900	1353	18540	2168	0.868	25.5	48.4	2330		
278	413.71	3.15	853100	21710	40.2	6.41	16728	1428	19571	2298	0.868	24.2	51.1	2713		
294	437.52	3.17	909900	23450	40.4	6.49	17740	1536	20760	2469	0.868	23.1	56.0	3185		
327	486.63	3.19	1021000	26730	40.6	6.57	19720	1733	23200	2800	0.867	21.1	64.4	4299		
331	492.58	3.19	1028000	26820	40.4	6.53	19845	1736	23413	2818	0.867	20.8	64.0	4433		
359	534.25	3.22	1131000	29970	40.8	6.63	21620	1925	25570	3126	0.866	19.5	73.0	5576		
392	583.36	3.24	1246100	33430	40.9	6.70	23600	2130	28039	3475	0.869	17.9	81.2	7230		
199	296.14	3.48	618700	28850	40.5	8.75	12600	1443	14220	2235	0.878	38.0	65.9	762.6	**40 x 16**	
215	319.95	3.50	696400	33120	41.3	9.00	14070	1656	15800	2555	0.887	34.3	76.0	1021		
249	370.55	3.51	812100	38480	41.5	9.03	16240	1924	18330	2976	0.887	29.9	89.4	1575		
277	412.22	3.53	909800	43410	41.7	9.10	18050	2160	20440	3348	0.887	27.1	101.5	2134		
297	441.98	3.53	966510	45500	41.4	8.98	19101	2264	21777	3529	0.882	25.8	106.7	2545		
324	482.16	3.55	1067480	50710	41.7	9.08	20931	2510	23923	3919	0.884	23.7	119.9	3311		
362	538.71	3.58	1202540	57630	41.8	9.16	23350	2832	26824	4436	0.883	21.5	137.6	4546		
397	590.80	3.60	1331040	64010	42.1	9.22	25597	3130	29530	4916	0.883	19.8	154.3	5927		
431	641.40	3.62	1450590	70280	42.1	9.27	27683	3412	32097	5379	0.882	18.5	170.7	7440		
503	748.55	3.67	1731940	85110	42.6	9.45	32433	4082	37881	6459	0.882	16.1	210.7	11670		
593	882.48	3.74	2096420	104970	43.2	9.66	38396	4952	45265	7874	0.883	14.0	265.7	18750		
230	342.28	3.71	867400	33120	44.6	8.71	15920	1656	18060	2568	0.876	38.9	92.7	1037	**44 x 16**	**W44**
262	389.90	3.73	1005000	38480	45.0	8.80	18280	1924	20780	2988	0.878	33.9	108.7	1564		
290	431.57	3.75	1126000	43410	45.2	8.87	20320	2160	23160	3362	0.879	30.7	123.5	2130		
335	498.53	3.77	1294000	49980	45.1	8.87	23150	2468	26600	3870	0.876	27.3	143.4	3135		

Note: Calculated based on 1 kg = 2.2046 lb and 1 m = 3.2808 feet.
*Please contact us for further clarification.